

Polynomials and Partial Fractions

Polynomials and rational functions are key components of algebra and calculus, playing a crucial role in various mathematical applications. Polynomials, are expressions involving non-negative integer powers of variables, date back to ancient Babylon and have been essential for solving equations. Rational functions, which are ratios of polynomials, allow for the exploration of complex relationships and behaviors, particularly in calculus, where they are used to study limits, derivatives, and integrals. Together, they underpin many principles in mathematics, physics, and engineering.

1 Polynomials

1.1 Definitions and Notation

In this subsection, we shall give basic definitions, notation and vocabulary of polynomials.

Let $\mathbb{K}$ be a field (Generally, we consider a field of characteristic 0 unless stated otherwise).

- We denote by $\mathbb{K}^{(\mathbb{N}_0)}$ the set of sequences $(a_n)_n$ with elements in $\mathbb{K}$, such that:

$$\exists p \in \mathbb{N}, \forall n \geq p, a_n = 0.$$

- An element of $\mathbb{K}^{(\mathbb{N}_0)}$ is called *polynomial*.
- Sometimes an element $(a_n)_n$ of $\mathbb{K}^{(\mathbb{N}_0)}$ is represented as $(a_0, a_1, a_2, \dots)$.
- We define on $\mathbb{K}^{(\mathbb{N}_0)}$ two binary operations: an addition and a multiplication as follows:

$$(a_n)_n + (b_n)_n = (a_n + b_n)_n$$

and

$$(a_n)_n (b_n)_n = (c_n)_n \quad \text{with} \quad c_n = \sum_{k=0}^n a_k b_{n-k}.$$

It is easy to verify that $(\mathbb{K}^{(\mathbb{N}_0)}, +, \cdot)$ is a commutative ring.

- The neutral element for addition is the polynomial $(0, 0, 0, \dots)$ called the *zero polynomial*.
- The neutral element for multiplication is the polynomial $(1, 0, 0, \dots)$.

- An element of $\mathbb{K}^{(\mathbb{N}_0)}$ of the form

$$(a, 0, 0, \dots)$$

is called a *constant polynomial*.

The map

$$\begin{aligned} f: \mathbb{K} &\rightarrow \mathbb{K}^{(\mathbb{N}_0)} \\ a &\mapsto (a, 0, 0, \dots) \end{aligned}$$

is a monomorphism of rings which allows to identify $(a_0, 0, 0, \dots)$ with $a_0 \in \mathbb{K}$ and so $\mathbb{K}$ can be considered as subring of $\mathbb{K}^{(\mathbb{N}_0)}$. Then, if $a \in \mathbb{K}$ and $(a_0, a_1, \dots, a_m, \dots) \in \mathbb{K}^{(\mathbb{N}_0)}$ we have

$$a(a_0, a_1, \dots, a_m, \dots) = (aa_0, aa_1, \dots, aa_m, \dots).$$

- We denote by X the polynomial

$$X = (0, 1, 0, 0, \dots).$$

With the convention $X^0 = (1, 0, 0, \dots)$, direct calculations give

$$X^2 = (0, 0, 1, 0, \dots), \quad X^3 = (0, 0, 0, 1, 0, \dots) \dots$$

Then we prove by induction on $k \in \mathbb{N}_0$ that

$$X^k = (0, \dots, 0, \underset{\substack{\uparrow \\ (k+1)th \text{ position}}}{1}, 0, 0, \dots).$$

Let P be an element of $\mathbb{K}^{(\mathbb{N}_0)}$:

$$P = (a_0, a_1, \dots, a_m, 0, 0, \dots) \text{ with } a_m \neq 0.$$

Then P can be written as

$$P = (a_0, 0, \dots) + (a_1, 0, \dots)X + (a_2, 0, \dots)X^2 + \dots + (a_m, 0, \dots)X^m,$$

which gives the following canonical writing of a polynomial:

$$P = a_0 + a_1X + a_2X^2 + \dots + a_mX^m = \sum_{k=0}^m a_kX^k.$$

- By definition, two polynomials $P = \sum_{i \geq 0} a_iX^i$ and $Q = \sum_{i \geq 0} b_iX^i$ are equal if and only if $a_n = b_n$ for all $n \in \mathbb{N}_0$.
- The ring $(\mathbb{K}^{(\mathbb{N}_0)}, +, \cdot)$ will be denoted $\mathbb{K}[X]$. The element X is called the *indeterminate*, and when $P = \sum_{k=0}^m a_kX^k$, the elements a_k ($1 \leq k \leq m$) are called *the coefficients* of P .

- The elements of $\mathbb{K}[X]$ are denoted by capital letters

$$A, B, C, P, Q, R, \dots$$

- Let $P \in \mathbb{K}[X]$.

- The *degree* of a nonzero polynomial P , denoted by $\deg P$, is defined by:

$$\deg P = \max\{n \in \mathbb{N}_0 : a_n \neq 0\}.$$

When $\deg P = p$, the coefficient a_p is called the *leading* coefficient of P . The polynomial P is said to be *monic* if $a_p = 1$, and a_0 is called the *constant term* of P .

- By convention the degree of the zero polynomial is $-\infty$.
- The *valuation* of a nonzero polynomial P is defined by:

$$\text{val}(P) = \min\{n \in \mathbb{N}_0 : a_n \neq 0\}.$$

- By convention the valuation of the zero polynomial is $+\infty$.

We denote by $\mathbb{K}_n[X]$ the set of polynomials in $\mathbb{K}[X]$ of degree less than or equal to n . It is easy to prove that $\mathbb{K}_n[X]$ is a subgroup of the group $(\mathbb{K}[X], +)$.

Example 1

1. In $\mathbb{R}[X]$, let $P = 2 + 3X^2 - 4X^3$. Then $\deg P = 3$, $\text{val} P = 0$, the leading coefficient of P is -4 , and the constant term of P is 2 .
2. Any nonzero constant polynomial is of degree zero. Conversely, any polynomial of degree zero is constant.
3. In $\mathbb{R}[X]$, let

$$P = (X + 1)^n - (X - 1)^n, n \geq 1,$$

Then $\deg P = n - 1$, the leading coefficient of P is $2n$, and the constant term of P is $1 - (-1)^n$.

We now give some properties of the degree. For that, we agree to the convention: $-\infty + -\infty = -\infty$, $a + -\infty = -\infty$, and $-\infty < a$ for any $a \in \mathbb{Z}$.

Proposition 2 For any $P, Q \in \mathbb{K}[X]$ and $\lambda \in \mathbb{K}^*$ we have:

1. $\deg(P + Q) \leq \max(\deg P, \deg Q)$,
2. $\deg(PQ) = \deg P + \deg Q$,

3. $\deg(\lambda P) = \deg P$.

Proof.

1. If P or Q is the zero polynomial, the inequality is obvious. Otherwise, let d be the degree of P and e the degree of Q . Then $P = a_0 + \cdots + a_d X^d$ and $Q = b_0 + \cdots + b_e X^e$, where a_i ($1 \leq i \leq d$) and b_j ($1 \leq j \leq e$), are in $\mathbb{K}$. If $d > e$, then we can write:

$$P + Q = (a_0 + b_0) + \cdots + (a_e + b_e)X^e + (a_{e+1})X^{e+1} + \cdots + a_d X^d.$$

It is then clear that

$$\deg(P + Q) = d = \max(\deg P, \deg Q).$$

The case when $d < e$ is similar. Finally, when $d = e$, we have:

$$P + Q = (a_0 + b_0) + \cdots + (a_d + b_d)X^d.$$

Either all coefficients are zero, and $\deg(P + Q) = -\infty$ making the inequality obvious, or at least one coefficient is nonzero and the nonzero coefficient with the highest index is the leading coefficient of $P + Q$, and its index is at most d .

2. We have

$$PQ = \sum_{k=0}^{n+m} c_k X^k,$$

where for any $0 \leq k \leq n + m$,

$$c_k = \sum_{i=0}^k a_i b_{k-i}.$$

Then,

$$c_{n+m} = a_n a_m \neq 0,$$

so c_{n+m} is the leading coefficient of PQ (since a_n and a_m are not zero).

Thus,

$$\deg(PQ) = \deg P + \deg Q.$$

3. If $\lambda \neq 0$ and $\deg P = n$, the leading coefficient of λP is λa_n , which is different from zero since $\lambda \neq 0$ and $a_n \neq 0$. Thus, $\deg(\lambda P) = n$.

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To give some properties of valuation, we agree to the convention: $+\infty + \infty = +\infty$, $a + \infty = +\infty$, $a < \infty$ for any $a \in \mathbb{Z}$.

Proposition 3 For any $P, Q \in \mathbb{K}[X]$, and $\lambda \in \mathbb{K}^*$, we have:

1. $\text{val}(P + Q) \geq \min(\text{val} P, \text{val} Q)$,
2. $\text{val}(PQ) = \text{val} P + \text{val} Q$,
3. $\text{val}(\lambda P) = \text{val} P$.

Proof.

1. It is easy to observe that if P or Q is the zero polynomial, we have the equality

$$\text{val}(P + Q) = \min(\text{val} P, \text{val} Q).$$

Suppose neither P nor Q is the zero polynomial, and let $m = \text{val} P$ and $\ell = \text{val} Q$. We can write

$$P = a_m X^m + a_{m+1} X^{m+1} + \cdots + a_d X^d \quad (a_m \neq 0)$$

and

$$Q = b_\ell X^\ell + b_{\ell+1} X^{\ell+1} + \cdots + b_e X^e \quad (b_\ell \neq 0).$$

If $\ell < m$ we have

$$P + Q = a_\ell X^\ell + a_{\ell+1} X^{\ell+1} + \cdots$$

Then, it is clear that $\text{val}(P + Q) = \ell = \min(\text{val} P, \text{val} Q)$. The case $m < \ell$ is similar. Finally, when $m = \ell$ we have

$$P + Q = (a_m + b_m) X^m + (a_{m+1} + b_{m+1}) X^{m+1} + \cdots$$

Either all coefficients are zero, and $\text{val}(P + Q) = +\infty$, making the inequality obvious, or at least one coefficient is nonzero. If the nonzero coefficient with the smallest index is $a_t + b_t$, then $\text{val}(P + Q) = t$, which is greater than or equal to m .

2. If P or Q is the zero polynomial, the desired equality is obvious. Suppose neither polynomial is zero. Let m be the valuation of P and ℓ the valuation of Q . Then

$$P = a_m X^m + \cdots + a_d X^d \quad (a_m \neq 0)$$

and

$$Q = b_\ell X^\ell + \cdots + b_e X^e \quad (b_\ell \neq 0).$$

Therefore, we can write:

$$PQ = a_m b_\ell X^{m+\ell} + \cdots + a_d b_e X^{d+e}$$

with $a_m b_\ell \neq 0$. Thus,

$$\text{val}(PQ) = m + \ell = \text{val} P + \text{val} Q.$$

3. Let $\lambda \neq 0$ and $P = a_0 + \cdots + a_d X^d$ with $\text{val} P = m$. This means that m is the least $n \in \mathbb{N}_0$ such that $a_n \neq 0$. Thus,

$$P = a_m X^m + a_{m+1} X^{m+1} + \cdots + a_d X^d$$

and $a_m \neq 0$. Since $\lambda \neq 0$, we have

$$\lambda P = \lambda a_m X^m + \lambda a_{m+1} X^{m+1} + \cdots + \lambda a_d X^d,$$

with $\lambda a_m \neq 0$. So

$$\text{val}(\lambda P) = m.$$

■

Remark 4 Let P be a nonzero polynomial in $\mathbb{K}[X]$. The following properties are obvious:

1. $\text{val} P \leq \deg P$,
2. $\text{val} 0 \geq \deg 0$,
3. If P is a constant polynomial, then $\text{val}(P) = \deg(P)$.

Let $P = \sum_{k=0}^n a_k X^k \in \mathbb{K}[X]$ ($n \in \mathbb{N}$). The map

$$\begin{aligned} \tilde{P} : \quad & \mathbb{K} \rightarrow \mathbb{K} \\ t \mapsto & \sum_{k=0}^n a_k t^k, \end{aligned}$$

(with the convention: $\forall t \in \mathbb{K}; t^0 = 1$) is called the *polynomial map* (function) associated with P . It is clear that for all $P, Q \in \mathbb{K}[X]$ and $\lambda \in \mathbb{K}$, we have:

1. $\widetilde{P+Q} = \tilde{P} + \tilde{Q}$.
2. $\widetilde{PQ} = \tilde{P}\tilde{Q}$.
3. $\widetilde{\lambda P} = \lambda \tilde{P}$.

► Let $P, Q \in \mathbb{K}[X]$ such that for any $t \in \mathbb{K}$ we have $\tilde{P}(t) = \tilde{Q}(t)$. This gives $(\widetilde{P-Q})(t) = 0, \forall t \in \mathbb{K}$. When $\mathbb{K}$ is infinite then, necessarily $(P-Q) = 0$ and so $P = Q$. This allows us, when the field $\mathbb{K}$ is infinite (e.g., when $\mathbb{K}$ is a subfield of $\mathbb{C}$), to identify the polynomial P with the polynomial map $\tilde{P}$. In the next example, we will show that this is not the case when $\mathbb{K}$ is finite.

Example 5

1. The polynomial map associated with the polynomial $P = 1 + X^3 \in \mathbb{R}[X]$ is the map from $\mathbb{R}$ into $\mathbb{R}$ defined by $\tilde{P}(t) = 1 + t^3$.

2. Let $P = -X + X^3 \in \mathbb{F}_3[X]$. We have $\tilde{P}(0) = \tilde{P}(1) = \tilde{P}(2) = 0$, and so P is identically zero as a map, But it is not the zero polynomial.

Theorem 6 The ring $\mathbb{K}[X]$ is an integral domain, and its only invertible elements are nonzero constant polynomials.

Proof. Let $P \in \mathbb{K}[X]$ be an invertible polynomial. Then there exists $Q \in \mathbb{K}[X]$ such that $PQ = 1$. Note that neither P nor Q can be zero. Taking the degree, we obtain

$$\deg(PQ) = \deg P + \deg Q = \deg 1 = 0.$$

Since $\deg P$ and $\deg Q$ are nonnegative integers and their sum is zero

$$\deg P = \deg Q = 0.$$

This is equivalent to say P and Q are nonzero constant polynomials. ■

1.2 Composition of Polynomials

Let P and Q be two polynomials of $\mathbb{K}[X]$ with

$$P = \sum_{i=0}^n a_i X^i.$$

We denote by $P \circ Q$ the polynomial defined as

$$P \circ Q = \sum_{i=0}^n a_i Q^i,$$

where it is always agreed that $Q^0 = 1$. The polynomial $P \circ Q$ is referred to as the *polynomial composed* of P and Q . It is denoted $P \circ Q$ or $P(Q)$ and is said to have been obtained by substitution of the polynomial Q for the indeterminate X in $P(X)$.

Example 7 In $\mathbb{R}[X]$, let $P = X^3 + X^2 + X + 1$ and $Q = 2X^2 + 1$. We have

1. $P(Q) = (2X^2 + 1)^3 + (2X^2 + 1)^2 + (2X^2 + 1) + 1 = 8X^6 + 20X^4 + 8X^2 + 4$.
2. $Q(P) = 2(X^3 + X^2 + X + 1)^2 + 1 = 2X^6 + 4X^5 + 6X^4 + 8X^3 + 6X^2 + 4X + 3$.

Proposition 8 Let $P, Q, R \in \mathbb{K}[X]$, and $\lambda \in \mathbb{K}$. We have

1. The composition of polynomials is not commutative,
2. Composition of polynomials is associative,
3. $X \circ P = P \circ X = P$,
4. $(PQ) \circ R = (P \circ Q)(Q \circ R)$,
5. $(P + \lambda Q) \circ R = (P \circ R) + (\lambda Q \circ R)$.
6. If P and Q are different from the zero polynomial we have $\deg(P \circ Q) = \deg P \deg Q$.

► The point 3) allows us to use either $P(X)$ or simply P .

1.3 Euclidean Algorithm

Theorem 9 For any $A, B \in \mathbb{K}[X]$ with $B \neq 0$, there exists a unique pair $(Q, R) \in \mathbb{K}[X]^2$ such that

$$A = BQ + R \quad \text{with} \quad \deg R < \deg B. \quad (1)$$

The polynomials Q and R are called the *quotient* and *remainder* of the Euclidean division (Long Division) of A by B , respectively. The polynomial B is said to divide A , or A is said to be divisible by B if $R = 0$.

Proof. ■

Existence: We prove the result by induction on the degree of A . When $\deg A < \deg B$, we take $(Q, R) = (0, A)$. Assume that $\deg A \geq \deg B$ and suppose that the property is true for any polynomial with degree is less than n . We show that it is true for a polynomial A with $\deg A = n + 1$. Consider

$$A_1 = A - \frac{a_{n+1}}{b_m} X^{n+1-m} B,$$

where a_{n+1} and b_m are the leading coefficients of A and B , respectively. Then $\deg(A_1) \leq n$. Applying the induction hypothesis to A_1 , there exists a pair (Q_1, R_1) of polynomials in $\mathbb{K}[X]$ such that

$$A_1 = Q_1 B + R_1 \quad \text{with} \quad \deg R_1 < \deg B,$$

which implies $A = \left(Q_1 + \frac{a_{n+1}}{b_m} X^{n+1-m}\right) B + R_1$.

Uniqueness: Suppose that (Q_1, R_1) and (Q_2, R_2) are two pairs of polynomials satisfying (1). Then

$$R_2 - R_1 = B(Q_1 - Q_2).$$

Consequently,

$$\deg(R_2 - R_1) = \deg(B(Q_1 - Q_2)) = \deg B + \deg(Q_1 - Q_2).$$

Since $\deg(R_2 - R_1) < \deg B$, necessarily $Q_1 - Q_2 = 0$ and $R_2 - R_1 = 0$, which gives $Q_1 = Q_2$ and $R_2 = R_1$.

Example 10

1. Every polynomial divides the zero polynomial.
2. $X^2 + X + 1$ divides $X^6 - 1$ since $X^6 - 1 = (X^2 + X + 1)(X^4 - X^3 + X + 1) + 0$.
3. For any $n \in \mathbb{N}_0$, $X - 1$ divides $X^n - 1$ since $X^n - 1 = (X - 1)(X^{n-1} + X^{n-2} + \dots + 1) + 0$.

Proposition 11

1. If $\lambda \in \mathbb{K}^*$, then λ divides any $A \in \mathbb{K}[X]$.

2. Let $A, B \in \mathbb{K}[X]$ and $B \neq 0$. If B divides A , then $\deg B \leq \deg A$.

Proof.

1. Since $\lambda \in \mathbb{K}^*$, we have $P = \lambda(\frac{1}{\lambda}P)$, so λ divides A as required.
2. If B divides A , we have $A = BQ$. Then

$$\deg A = \deg B + \deg Q.$$

Hence, $\deg B \leq \deg A$.

■

Example 12 In $\mathbb{R}[X]$, let

$$A = 4X^5 - 3X^4 + 5X^2 + 1$$

and

$$B = X^3 + 1.$$

We have

$$\begin{array}{r|l} 4X^5 - 3X^4 + 5X^2 + 1 & \frac{X^3 + 1}{4X^2 - 3X} \\ \hline 4X^5 + 4X^2 & \underbrace{4X^2 - 3X}_Q \\ \hline -3X^4 + X^2 + 1 & \\ \hline -3X^4 - 3X & \\ \hline +X^2 + 3X + 1 & \\ \hline \underbrace{}_R & \end{array}$$

Then

$$\underbrace{4X^5 - 3X^4 + 5X^2 + 1}_A = \underbrace{(X^3 + 1)}_B \underbrace{(4X^2 - 3X)}_Q + \underbrace{X^2 + 3X + 1}_R.$$